```
mysql> create database assignment2;
Query OK, 1 row affected (0.07 sec)

mysql> use assignment2;
Database changed
mysql> create table department(dept_id int primary key, dept_name varchar(30), city varchar(20));
Query OK, 0 rows affected (0.21 sec)

mysql> create table employee(emp_id int primary key, name varchar(40) Not null, salary decimal, joining_date date,dept_id int , foreign key (dept_id) refere
nces department(dept_id));
Query OK, 0 rows affected (0.07 sec)

mysql> desc department;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| dept_id   | int         | NO   | PRI | NULL    |       |
| dept_name | varchar(30) | YES  |     | NULL    |       |
| city      | varchar(20) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.15 sec)

mysql> desc employee;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| emp_id      | int           | NO   | PRI | NULL    |       |
| name        | varchar(40)   | NO   |     | NULL    |       |
| salary      | decimal(10,0) | YES  |     | NULL    |       |
| joining_date | date          | YES  |     | NULL    |       |
| dept_id     | int           | YES  | MUL | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

```
mysql> insert into department(dept_id, dept_name, city) values (10, 'IT', 'Pune'),
    -> (20, 'HR', 'Mumbai'),
    -> (30, 'Sales', 'Mumbai'),
    -> (40, 'Finance', 'Delhi'),
    -> (50, 'Marketing', 'Nashik');
Query OK, 5 rows affected (0.07 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> insert into table (emp_id, name, salary, joining_date, dept_id) values(101, 'Amit', 60000, '2022-01-10', 10),
    -> (102, 'Priya', 80000, '2021-06-15', 10),
    -> (103, 'Rahul', 75000, '2023-03-20', 30),
    -> (104, 'Sneha', 50000, '2022-08-01', 20),
    -> (105, 'Neha', 60000, '2021-04-12', 30),
    -> (106, 'Karan', 45000, '2023-01-05', 20),
    -> (107, 'Pooja', 55000, '2022-11-18', 30),
    -> (108, 'Vikram', 55000, '2021-09-22', 10),
    -> (109, 'Meera', 40000, '2024-02-14', 10),
    -> (110, 'Arjun', 50000, '2023-07-09', 20);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
able (emp_id, name, salary, joining_date, dept_id) values(101, 'Amit', 60000, '' at line 1
mysql> insert into employee (emp_id, name, salary, joining_date, dept_id) values(101, 'Amit', 60000, '2022-01-10', 10),
    -> (102, 'Priya', 80000, '2021-06-15', 10),
    -> (103, 'Rahul', 75000, '2023-03-20', 30),
    -> (104, 'Sneha', 50000, '2022-08-01', 20),
    -> (105, 'Neha', 60000, '2021-04-12', 30),
    -> (106, 'Karan', 45000, '2023-01-05', 20),
    -> (107, 'Pooja', 55000, '2022-11-18', 30),
    -> (108, 'Vikram', 55000, '2021-09-22', 10),
    -> (109, 'Meera', 40000, '2024-02-14', 10),
    -> (110, 'Arjun', 50000, '2023-07-09', 20);
Query OK, 10 rows affected (0.07 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+-----------+--------+
| dept_id | dept_name | city   |
+---------+-----------+--------+
|      10 | IT        | Pune   |
|      20 | HR        | Mumbai |
|      30 | Sales     | Mumbai |
|      40 | Finance   | Delhi  |
|      50 | Marketing | Nashik |
+---------+-----------+--------+
5 rows in set (0.00 sec)

mysql> select * from employee;
+---------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+---------+--------+--------+--------------+---------+
|    101 | Amit   |  60000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  60000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  55000 | 2021-09-22   |      10 |
|    109 | Meera  |  40000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
+---------+--------+--------+--------------+---------+
10 rows in set (0.00 sec)
```

```
mysql> update employee set salary = 40000 where emp_id =105;
Query OK, 1 row affected (0.07 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from employee;
+--------+--------+--------+-------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+-------------+---------+
|    101 | Amit   |  60000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  55000 | 2021-09-22   |      10 |
|    109 | Meera  |  40000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
+--------+--------+--------+-------------+---------+
10 rows in set (0.00 sec)

mysql> grant select on assignment2.employee
    -> to 'readonly_user'@'localhost';
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'readonly_user'@'localhost'
    -> IDENTIFIED BY 'Read@123';
Query OK, 0 rows affected (0.11 sec)

mysql> grant select on assignment2.employee
    -> to 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE USER 'temp_user'@'localhost'
    -> IDENTIFIED BY 'Temp@123';
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> GRANT INSERT ON assignment2.employee
    -> TO 'temp_user'@'localhost';
Query OK, 0 rows affected (0.07 sec)

mysql> revoke insert on assignment2.employee
    -> from 'temp_user'@'localhost';
Query OK, 0 rows affected (0.07 sec)

mysql> select * from employee;
+--------+--------+--------+-------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+-------------+---------+
|    101 | Amit   |  60000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  55000 | 2021-09-22   |      10 |
|    109 | Meera  |  40000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
+--------+--------+--------+-------------+---------+
10 rows in set (0.00 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> insert into employee(emp_id, name, salary, joining_date, dept_id) values (116, 'Anjali', 72000, '2024-06-10', 10);
Query OK, 1 row affected (0.07 sec)

mysql> select * from employee;
+--------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+--------------+---------+
|    101 | Amit   |  60000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  55000 | 2021-09-22   |      10 |
|    109 | Meera  |  40000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
|    116 | Anjali |  72000 | 2024-06-10   |      10 |
+--------+--------+--------+--------------+---------+
11 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.07 sec)

mysql> savepoint b1;
Query OK, 0 rows affected (0.00 sec)

mysql> update employee set salary = 75000 where emp_id =101;
Query OK, 1 row affected (0.07 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from employee;
+--------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+--------------+---------+
|    101 | Amit   |  75000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
```

```
mysql> select * from employee where salary > 50000;
+--------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+--------------+---------+
|    101 | Amit   |  75000 | 2022-01-10   |      10 |
|    102 | Priya  |  80000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  55000 | 2021-09-22   |      10 |
|    116 | Anjali |  72000 | 2024-06-10   |      10 |
+--------+--------+--------+--------------+---------+
6 rows in set (0.07 sec)

mysql> update employee set salary = salary *1.10
    -> where dept_id = 10;
Query OK, 5 rows affected (0.08 sec)
Rows matched: 5  Changed: 5  Warnings: 0

mysql> select * from employee;
+--------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+--------------+---------+
|    101 | Amit   |  82500 | 2022-01-10   |      10 |
|    102 | Priya  |  88000 | 2021-06-15   |      10 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    108 | Vikram |  60500 | 2021-09-22   |      10 |
|    109 | Meera  |  44000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
|    116 | Anjali |  79200 | 2024-06-10   |      10 |
+--------+--------+--------+--------------+---------+
11 rows in set (0.00 sec)

mysql> create view mumbai_salesperson as
    -> select e.emp_id, e.name,d.dept_id, d.dept_name, d.city from employee as e
    -> join department as d on e.dept_id = d.dept_id
    -> where d.dept_name = 'sales' and d.city = 'mumbai';
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> select * from mumbai_salesperson;
+--------+-------+---------+-----------+--------+
| emp_id | name  | dept_id | dept_name | city   |
+--------+-------+---------+-----------+--------+
|    103 | Rahul |      30 | Sales     | Mumbai |
|    105 | Neha  |      30 | Sales     | Mumbai |
|    107 | Pooja |      30 | Sales     | Mumbai |
+--------+-------+---------+-----------+--------+
3 rows in set (0.07 sec)

mysql> select dept_id, count(*) as totalEmployee from employee ;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'assignment2.employee.dept_id'; this is
 incompatible with sql_mode=only_full_group_by
mysql> select dept_id, count(*) as totalEmployee from employee group by dept_id;
+---------+---------------+
| dept_id | totalEmployee |
+---------+---------------+
|      10 |             5 |
|      20 |             3 |
|      30 |             3 |
+---------+---------------+
3 rows in set (0.04 sec)

mysql> select max(salary)as highest_salary from employee;
+----------------+
| highest_salary |
+----------------+
|          88000 |
+----------------+
1 row in set (0.07 sec)

mysql> select distinct(salary) from employee order by salary desc limit 1 offset 1;
+--------+
| salary |
+--------+
|  82500 |
+--------+
1 row in set (0.07 sec)
```

```
mysql> select * from employee order by joining_date desc;
+--------+--------+--------+--------------+---------+
| emp_id | name   | salary | joining_date | dept_id |
+--------+--------+--------+--------------+---------+
|    116 | Anjali |  79200 | 2024-06-10   |      10 |
|    109 | Meera  |  44000 | 2024-02-14   |      10 |
|    110 | Arjun  |  50000 | 2023-07-09   |      20 |
|    103 | Rahul  |  75000 | 2023-03-20   |      30 |
|    106 | Karan  |  45000 | 2023-01-05   |      20 |
|    107 | Pooja  |  55000 | 2022-11-18   |      30 |
|    104 | Sneha  |  50000 | 2022-08-01   |      20 |
|    101 | Amit   |  82500 | 2022-01-10   |      10 |
|    108 | Vikram |  60500 | 2021-09-22   |      10 |
|    102 | Priya  |  88000 | 2021-06-15   |      10 |
|    105 | Neha   |  40000 | 2021-04-12   |      30 |
+--------+--------+--------+--------------+---------+
11 rows in set (0.00 sec)

mysql> alter table department
    -> rename column dept_name to  DepartmentName;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+----------------+--------+
| dept_id | DepartmentName | city   |
+---------+----------------+--------+
|      10 | IT             | Pune   |
|      20 | HR             | Mumbai |
|      30 | Sales          | Mumbai |
|      40 | Finance        | Delhi  |
|      50 | Marketing      | Nashik |
+---------+----------------+--------+
5 rows in set (0.00 sec)
```